

CAPE NATURALISTE, Australia

WMO: 946000

Lat: 33.5372S

Lon: 115.0189E

Elev: 110

StdP: 100.02

Time Zone: 8.00 (E08)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.5	8.4	2.5	4.6	11.0	3.3	4.9	11.4	18.2	12.7	16.5	13.4	5.8	130	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.0	30.8	19.2	28.9	18.8	27.2	18.5	20.9	27.0	20.1	26.1	19.5	25.2	5.5	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.0	22.9	18.1	13.2	22.4	17.4	12.6	22.1	60.9	26.9	58.2	26.3	56.0	25.2	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.4	13.8	12.4	5.9	35.2	0.9	1.3	5.2	36.2	4.7	37.0	4.2	37.7	3.6	38.7		
(5)			4.2	22.3	0.8	1.0	3.6	23.0	3.1	23.6	2.6	24.1	2.0	24.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.9	20.6	21.0	20.0	18.2	16.3	14.7	13.6	13.5	14.0	15.3	17.5	19.0	(6)
(7)		DBStd	3.47	2.67	2.64	2.62	2.11	1.96	1.90	1.85	1.79	1.88	1.95	2.41	2.86	(7)
(8)		HDD10.0	2	0	0	0	0	0	0	1	1	1	0	0	0	(8)
(9)		HDD18.3	813	6	4	11	28	67	108	147	148	131	97	43	23	(9)
(10)		CDD10.0	2536	328	307	308	244	196	142	112	111	119	164	226	278	(10)
(11)		CDD18.3	308	75	78	61	23	5	1	0	0	0	3	19	43	(11)
(12)		CDH23.3	1862	521	507	339	49	2	0	0	0	0	12	119	313	(12)
(13)		CDH26.7	483	141	133	84	3	0	0	0	0	0	1	24	98	(13)
(14)	Wind	WSAvg	7.1	7.4	7.3	6.9	6.3	6.6	7.5	7.7	7.2	7.3	7.0	7.1	7.3	(14)
(15)	Precipitation	PrecAvg	834	12	11	25	46	116	170	172	115	80	50	29	13	(15)
(16)		PrecMax	1109	58	58	154	188	240	311	338	211	170	143	100	57	(16)
(17)		PrecMin	584	0	0	1	0	46	69	99	50	29	5	0	0	(17)
(18)		PrecStd	143	17	15	33	32	48	58	53	42	30	27	23	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.4	32.3	32.4	26.7	23.2	20.7	20.0	21.2	25.2	29.3	32.8	(19)	
(20)			MCWB	19.8	20.0	19.3	17.8	17.5	15.9	16.2	15.5	15.7	16.2	17.3	18.5	(20)
(21)		2%	DB	30.0	30.2	28.9	24.7	21.8	19.5	18.4	18.6	19.5	22.4	26.5	29.6	(21)
(22)			MCWB	19.3	19.3	18.5	17.8	16.8	15.4	14.7	14.6	15.0	16.0	17.3	18.9	(22)
(23)		5%	DB	27.8	28.1	26.7	23.3	20.8	18.7	17.5	17.6	18.4	20.6	24.2	26.8	(23)
(24)			MCWB	18.8	19.0	18.1	17.1	16.1	14.9	14.0	13.9	14.2	15.3	16.6	18.0	(24)
(25)		10%	DB	26.2	26.4	24.9	22.1	19.8	17.9	16.7	16.7	17.3	19.3	22.3	24.4	(25)
(26)			MCWB	18.5	18.7	17.9	16.7	15.8	14.4	13.4	13.2	13.6	14.5	16.3	17.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.7	22.4	21.2	20.1	19.0	17.7	17.1	16.6	17.0	18.0	19.4	21.0	(27)
(28)			MCDWB	28.2	28.3	26.5	23.3	21.7	19.3	18.7	18.6	19.8	22.0	24.5	27.6	(28)
(29)		2%	WB	20.6	20.9	20.2	19.1	17.9	16.4	15.6	15.5	15.7	16.8	18.3	19.8	(29)
(30)			MCDWB	27.3	27.0	25.5	22.7	20.5	18.6	17.7	17.7	18.4	20.8	23.7	26.9	(30)
(31)		5%	WB	19.7	20.1	19.4	18.3	17.1	15.6	14.5	14.6	14.8	15.9	17.5	18.8	(31)
(32)			MCDWB	26.3	26.2	24.5	21.7	19.9	18.0	16.9	17.0	17.7	19.8	22.7	25.1	(32)
(33)		10%	WB	19.0	19.3	18.7	17.5	16.2	14.7	13.8	13.7	14.0	15.1	16.8	17.9	(33)
(34)			MCDWB	24.9	25.1	23.6	21.1	19.3	17.4	16.2	16.2	16.8	18.8	21.5	23.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.1	10.0	9.3	7.7	6.5	5.3	5.4	5.8	6.4	7.8	8.9	9.6	(35)
(36)			MCDBR	12.8	12.4	11.8	9.0	7.2	5.8	5.9	6.6	7.5	9.7	11.9	13.0	(36)
(37)			MCWBR	5.8	5.2	5.1	4.9	4.5	3.9	4.2	4.3	4.9	5.7	5.9	5.8	(37)
(38)		5% WB	MCDWB	10.8	10.2	9.3	7.5	6.3	5.3	5.5	5.9	6.8	8.6	10.0	11.2	(38)
(39)			MCWBR	5.6	5.1	5.0	4.9	4.8	4.2	4.6	4.5	5.0	5.6	5.7	5.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.330	0.330	0.320	0.315	0.310	0.305	0.303	0.312	0.324	0.323	0.326	0.323	(40)	
(41)		taud	2.622	2.635	2.661	2.669	2.646	2.643	2.647	2.611	2.559	2.579	2.593	2.625	(41)	
(42)		Ebn at Noon	1008	988	963	915	866	845	866	904	943	984	1006	1019	(42)	
(43)		Edh at Noon	101	96	88	78	70	66	69	81	95	101	103	101	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.88	7.03	5.62	4.01	2.91	2.33	2.49	3.35	4.51	5.98	7.20	7.89	(44)	
(45)		RadStd	0.40	0.29	0.29	0.24	0.19	0.15	0.12	0.20	0.24	0.30	0.40	0.36	(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page